MIL-V-24586A(SH)
14 August 1985
SUPERSEDING
MIL-V-24586(SH)
29 November 1978

MILITARY SPECIFICATION

# VALVE, NEEDLE, SIZE 1/4-INCH AND 1/2-INCH, UNION BONNET CONSTRUCTION

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 Scope. This specification covers the requirements of 1/4-inch and 1/2-inch sized needle valves.

* 1.2 Classification. Valves shall be of the sizes, classes, styles and types, as specified (see 6.2.1).

* 1.2.1 Classes. Classes shall be as follows:

Class I - Valve shall be rated for 1440 pounds per square inch ($lb/in^2$) minimum at a temperature range of minus 20 to 100 degrees Fahrenheit (°F), and 1240 $lb/in^2$ minimum at 200°F.
Class II - Valve shall be rated for 6000 $lb/in^2$ minimum at a temperature range of minus 20 to 100°F, and 5160 $lb/in^2$ minimum at 200°F.

* 1.2.2 Style. Styles shall be as follows:

Style A - Inline globe.
Style B - Angle globe.

* 1.2.3 Type. Types shall be as follows:

Type I - Stainless steel.
Type II - Carbon steel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges, (Except for Systems Indicated Herein); Packaging of.

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems Requirement for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-798 - Nondestructive Testing, Welding, Quality Control, Material Control and Identification and Hi-Shock Test Requirements for Piping System Components for Naval Shipboard Use.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street, NW, Washington, DC 20036.)



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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
B16.11 - Forged Steel Fittings, Socket-Welding and Threaded. (DoD adopted)
B46.1 - Surface Texture Surface Roughness, Waviness and Lay. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 105 - Standard Specification for Forgings, Carbon Steel, for Piping Components.
A 108 - Standard Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality. (DoD adopted)
A 182 - Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service. (DoD adopted)
A 194 - Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High-Temperature Service. (DoD adopted)
A 216 - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High Temperature Service. (DoD adopted)
A 473 - Standard Specification for Stainless and Heat-Resisting Steel Forgings. (DoD adopted)
A 582 - Standard Specification for Free-Machining and Heat-Resisting Steel Bars, Hot-Rolled or Cold-Finished. (DoD adopted)
A 743 - Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Nickel-Base, Corrosion Resistant, for General Application. (DoD adopted)

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Materials. Materials shall be as specified in table I.



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TABLE I. List of materials.

| Name of parts | Type I | | Type II | |
|---|---|---|---|---|
| | Material | Applicable documents | Material | Applicable documents |
| Body | Stainless steel | ASTM A 182, grade F316<br>ASTM A 743, grade CF-8M | Forged or cast steel | ASTM A 182, grade F1<br>ASTM A 105<br>ASTM A 216, grade WCB |
| Bonnet | | | | |
| Union nut | Stainless steel | ASTM A 194, grade B8MA, class 1B | ---- | ASTM A 105, ASTM A 194, ASTM A 108, grade 4 |
| Stem | Stainless steel | ASTM A 582, type 416<br>AISI 410, or 416<br>QQ-S-763, class 410<br>ASTM A 473, type 410 or 416 | Stainless steel | ASTM A 473, type 410, or 416<br>ASTM A 582, type 416<br>AISI 410, or 416<br>QQ-S-763, class 410 |
| Gland follower | Stainless steel | ASTM A 473, type 410, or 416 | ---- | ASTM A 182, grade F1<br>ASTM A 105, ASTM A 108 |
| Packing nut | | ASTM A 194, grade B8MA | | ASTM A 105, ASTM A 108<br>ASTM A 194, grade 4 |
| Packing | | | | |

3.1.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Design.

3.2.1 Bonnet construction. Valve shall be union bonnet construction.

3.2.2 Back seat. Valve shall have a metal-to-metal back seat.

3.2.3 End connections. Valve shall be socket welded with the socket inside dimensions in accordance with ANSI B16.11.

3.2.4 Yoke construction. Valve shall be an inside screw construction.

3.2.5 Main seat body. Valve seats shall be integral with the body.

3.2.6 Main seat disc. Valve disc seating surface may be integral with the stem.



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3.2.7 Packing assembly and nut. Packing gland follower and packing nut shall be separate. Packing nut shall be threaded to permit packing adjustment.

3.2.8 Body-to-bonnet assembly. Body and bonnet assembly shall be body guided.

3.2.8.1 Body-to-bonnet joint. Body and bonnet joint shall be a metal-to-metal surface with a finish of 63 maximum in accordance with ANSI B46.1, or shall incorporate a deformable gasket.

* 3.2.9 Dimensions. Maximum dimensions shall be in accordance with table II.

TABLE II. Maximum dimensions.

| Size | 1/4-inch | | 1/2-inch | |
|---|---|---|---|---|
| Style | A | B | A | B |
| End-to-end (inches) | 2-1/4 | — | 2-7/8 | — |
| Center-to-end (inches) | — | 1-1/8 | — | 1-1/2 |
| Height (inches) | 4 | 4 | 5-1/2 | 5-1/2 |

NOTE: Height is measured with valve in open position, from socket centerline to top of handwheel.

* 3.2.10 Flow coefficient. Valves shall be designed with a minimum flow coefficient ($C_v$) of 0.4 for 1/4-inch valves, and 1.5 for 1/2-inch valves.

* 3.2.11 Handwheel diameter and assembly. Handwheel shall be round, or of cross or tee-shape. Maximum diameter of handwheel shall be 3 inches. Handwheel shall be attached to the stem with a locknut to prevent loss of handwheel.

3.2.12 Body markings and identification information.

3.2.12.1 Body markings. Material identification shall be forged on the body.

* 3.2.12.2 Identification information. Identification information shall be attached to handwheel and stem by the handwheel locknut (see 3.2.11). As an alternative method, identification information may be attached to or permanently marked on the valve body. If this method is used, the caution of (d) below shall appear on both sides of the valve body. Any permanent marking on the valve body shall be forged, cast, low stress die-stamped, or electroetched. Attachment of identification information, in any case, shall not result in any loose parts and shall not interfere with valve operation. Identification information shall include the following:

(a) Manufacturers name and model number on identification plate.
(b) Pressure rating and size on body.
(c) Body material specification on body.
(d) Indicate on nameplate; CAUTION: Stem must be back seated against the bonnet prior to and during assembly of valve.



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3.2.13 Main seat leakage. Maximum permissible seat leakage rate shall be 10 milliliters (mL) per hour at rated pressure. The testing fluid shall be clean tap water except cutting oil may be present. Pressure shall be applied to the underside of the disc.

3.2.14 Shock. Valves furnished in accordance with this specification shall conform to grade A, class 1, type C of MIL-S-901 and MIL-STD-798.

* 3.2.14.1 Acceptance criteria for shock test shall be as follows:

(a) Momentary malfunction at the time of the impact blow is permitted and acceptable.
(b) Permanent deformation, misalignment, and functional impairments shall be cause for rejection.
(c) Permissible seat leakage shall be as specified in 3.2.13.

3.2.15 Threaded parts. Threaded parts shall be in accordance with ANSI B1.1.

3.2.16 Operating force. Valves shall open and close fully with a tangential force at the handwheel rim of 50 pounds or less.

* 3.2.17 Shell pressure. Class I valves shall be designed to withstand a minimum hydrostatic shell test pressure of 2175 $lb/in^2$. Class II valves shall withstand a minimum hydrostatic shell test pressure of 9000 $lb/in^2$.

3.2.18 Drawings. The contractor shall prepare drawings in accordance with the data ordering document included in the contract or order (see 6.2.2).

* 3.2.19 Repair parts. Valves shall be designed such that the only repair parts required are gaskets (if used) and packing. Valves shall be designed such that special tools are not required for installation and repair.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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* 4.1.2 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the contract or order (see 6.2.1).

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.3 and the tests specified in 4.4.

* 4.2.1 Lot. Valves of the same size, style and type (see 1.2) offered for delivery at the same time and manufactured within the preceding 12 months shall be considered a lot for purposes of sampling.

4.2.2 Sampling for visual and dimensional examination. Sampling of each lot for visual and dimensional examination specified in 4.3 shall be conducted in accordance with MIL-STD-105, general inspection level II, AQL 2.5 percent.

4.3 Visual and dimensional examination. Sample valves shall be visually and dimensionally examined to determine conformance with the specified design, inclusion of all required parts, proper assembly, the existence of proper markings and dimensions specified on the applicable specifications and standards.

4.3.1 The Defense Contract Administration Services Management Area (DCASMA) shall normally accept certification that parts conform to their applicable material specification providing the certification contains actual test results. Certifications representing stocks more than 1 year old are not acceptable.

4.4 Tests.

4.4.1 Shock. Valve design shall be shock tested in accordance with MIL-S-901 and MIL-STD-798 to assure compliance with the requirements specified in 3.2.14 and 3.2.14.1.

4.4.2 Shell pressure test. Each valve shall be hydrostatically tested to assure compliance with the requirements specified in 3.2.13 and 3.2.17.

* 4.4.3 Seat leakage test. Seat leakage testing shall be conducted to assure compliance with the requirements specified in 3.2.13.

4.5 Shock test report. The contractor shall prepare a shock test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

* 4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging, packing and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B or C and marked in accordance with MIL-V-3, as specified (see 6.2.1).



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5.1.1 Special marking. In addition to the marking required in 5.1, shelf life markings, where applicable (O-rings) shall be applied to repair parts packages containing nonmetallic items.

5.1.2 Cushioning, dunnage and wrapping materials.

5.1.2.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation and stowage.

5.1.2.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned with loose-fill material shall not be taken on board ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

5.1.2.3 Cushioning, filler, dunnage, and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

6. NOTES

* 6.1 Intended use. Needle valves covered by this specification are intended for use for vent, drain, and fluid sampling service. This specification is not intended to describe valves for use in steam systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Size, class, style and type of valve required (see 1.2).
(c) Specify inspection system (see 4.1.2).
(d) Levels of preservation-packaging and packing required (see 5.1).
(e) Allowance parts list or National stock number, if available.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.



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| Paragraph no. | Data requirements title | Applicable DID no. | Option |
|---|---|---|---|
| 3.2.18 | Drawings, engineering and associated lists | DI-E-7031 | Level 2<br>Design activity designation - Contractor<br>Drawing number - Contractor<br>Delivery of hard copies - Contracting activity |
| 4.5 | Report, equipment shock test | UDI-T-23753 | ——— |

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data where submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

* 6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

* 6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.



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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-V-24586A(SH) | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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